

Role of iron in neurotoxicity: a cause for concern in the elderly?



PURPOSE OF REVIEW: To explore the role of iron physiology in the brain of healthy adults and review how increased brain iron deposition has been associated with common neurodegenerative diseases that affect the elderly. RECENT FINDINGS: Because iron plays a role in oxygen transportation, myelin synthesis, neurotransmitter production, and electron transfers, it serves as a crucial cofactor in normal central nervous metabolism. However, an increased level of brain iron may promote neurotoxicity due to free radical formation, lipid peroxidation, and ultimately, cellular death. Advanced neuroimaging techniques and pathological studies have demonstrated increased brain iron with aging, and increased iron deposition has also been observed in patients with a constellation of neurological diseases, including Alzheimer's disease, Parkinson's disease, and stroke. SUMMARY: Pathologic and neurologic imaging coupled with experimentation have increased our understanding of the link between iron and neurodegeneration. A potential implication is that disease-modifying therapies aimed at removing excess iron may one day be part of the armamentarium employed by clinicians to decrease the burden of neurodegenerative diseases in the elderly.